

Applying Relation Extraction and Graph Matching to Answering Multiple Choice Questions

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Abstract

In this research, we combine Transformer-based relation extraction with matching of knowledge graphs (KGs) and apply them to answering multiple-choice questions (MCQs) while maintaining the traceability of the output process. KGs are structured representations of factual knowledge consisting of entities and relations. Due to the high construction cost, they had been regarded as static databases with validated links. However, the recent development of Transformer-based relation extraction (RE) methods has enabled us to generate KGs dynamically by giving them natural language texts, and thereby opened the possibility for representing the meaning of the input sentences with the created KGs. Using this effect, we propose a method that answers MCQs in the “fill-in-the-blank” format, taking care of the point that RE methods generate KGs that represent false information if provided with factually incorrect texts. We measure the truthfulness of each question sentence by (i) converting the sentence into a relational graph using an RE method and (ii) verifying it against factually correct KGs under the closed-world assumption. The experimental results demonstrate that our method correctly answers up to around 70% of the questions, while providing traceability of the procedure. We also highlight that the question category has a vast influence on the accuracy.

1 Introduction

In this research, we combine Transformer-based relation extraction and matching of **knowledge graphs (KGs)** and apply the combination to answer multiple-choice questions while keeping the output process traceable. To extract the relations of entities from a given natural language text is called **relation extraction (RE)**, and it is an important step to automatically construct KGs. Since the construction of KGs required massive human annotations, KGs have long been regarded as static databases with only factually correct links. However, the recent development of Transformer-based RE methods has enabled us to construct KGs dynamically for any text in a natural language, e.g., English, at a low cost. KGs have a graph structure, that is, a discrete data structure, and algorithms for graphs are applicable to them. In particular, deciding whether a KG matches another KG as a subgraph, which semantically means the latter KG entails the former, can be regarded as a graph matching problem. Therefore, we conjecture that entailment of one sentence in natural language by another could be checked by apply-

ing RE to the sentences and graph matching to the obtained KGs. Moreover, the matching of KGs makes the entailment relation between the sentences traceable. One of the tasks that needs such traceable entailment of natural language sentences is the automatic answering of multiple-choice questions.

Multiple-choice questions (MCQs) are widely used in higher education due to their objective evaluation and the ability to conduct exams efficiently on a large scale. On the learners’ side, receiving the reasons why each option is correct or incorrect after a test is effective for both enhancing the retention of the topic and reducing the misunderstanding of facts caused by reading the false options (Butler and Roediger 2008). However, on the teacher’s side, it is not easy to create explanations for all the answers. In order to reduce their burden, large language models (LLMs) might be helpful by asking them to generate step-by-step explanations leading to the answers. Unfortunately, in the current status, it is known that such explanations often lead to reasoning errors depending on the properties of the options in MCQs and the order of them (Turpin et al. 2023). Moreover, even though it is becoming less often, there remains a risk of hallucination, where non-existent facts are stated, and reasoning based on such hallucinations makes reliability a persistent issue.

Our study tries to overcome these problems by combining a Transformer-based RE method and a rule-based KG matching to propose a method for answering MCQs with reliable explanations. We use an RE method to create a tailored relational graph for representing the meaning of the question sentence of the MCQ. By measuring the truthfulness of the obtained graph and verifying it against factually correct KGs under the closed-world assumption, our method outputs the most reliable word from the given choices with its explanation.

This paper is organized as follows. First, in Section 2, we provide the formal definition of the task that we addressed in this study. Section 3 introduces the key concepts and methods that underpin our discussion. In Section 4, we describe the details of the proposed method, followed by Section 5, where we present the results of answering experiments conducted on original MCQ datasets. Section 6 reviews related works in the fields of multiple-choice question answering and fact verification. Finally, Section 7 concludes the paper

and outlines directions for future work.

2 Multiple Choice Question Answering

Multiple choice question answering (MCQA) is a task where a model is applied to select the correct option given a question sentence q , a set of options $\mathcal{O} = \{o_1, \dots, o_k\}$, and optional background knowledge $\mathcal{C} = \{C_1, \dots, C_k\}$ related to each choice o_i ($i = 1, \dots, k$) (Shah, Gupta, and Roth 2020). We treat the case of the 4-way cloze test format, where the task is to choose the best word that fits in the blank from four options. This problem is defined with a pair of a question sentence $q(x)$ with a blank x and a set of choices $\mathcal{O} = \{o_1, o_2, o_3, o_4\}$, where there is exactly one correct answer among o_i ($i = 1, 2, 3, 4$). An example of $q(x)$ and $\mathcal{O}$ is as follows.

Example 1 (Barack Obama). *Let $q(x)$ be “ $\{x\}$ is an American politician and attorney who served as the 44th president of the United States from 2009 to 2017,” where*

$$\mathcal{O} = \left\{ \begin{array}{l} o_1: \text{“Barack Obama”,} \\ o_2: \text{“Joe Biden”,} \\ o_3: \text{“George Walker Bush”,} \\ o_4: \text{“Bill Clinton”} \end{array} \right\}.$$

The correct answer is o_1 : “Barack Obama”.

In short, the MCQA that we tackle in this study is formalized as follows:

Definition 1 (Cloze test 4-way MCQA). *Cloze test 4-way MCQA is a task of selecting the correct option o_i given a question sentence $q(x)$ from the set of four option words $\mathcal{O} = \{o_1, o_2, o_3, o_4\}$ with a blank x and a set of background knowledge $\mathcal{C} = \{C_1, C_2, C_3, C_4\}$ corresponding to each o_i . The task is to find the correct answer $o_i \in \mathcal{O}$ such that*

$$C_i \models q(o_i), \quad (1)$$

where $\models$ informally indicates the entailment between natural language sentences.

Our requirement is to give a transformation $\mathcal{G}$ with which $q(x)$ and C_i are transformed into some objects $\mathcal{G}(q(x))$ and $\mathcal{G}(C_i)$, respectively, and to provide a **proof method** for deciding

$$\mathcal{G}(C_i) \vdash \mathcal{G}(q(o_i)),$$

to solve the entailment problem (1). We adopt the set of knowledge graphs (KGs) as the co-domain of $\mathcal{G}$ because a KG is regarded as a set of conjunctions of binary ground atoms in first-order logic. Then the method of $\vdash$ is the inclusion checking of two conjunctions, which can be seen as the subgraph isomorphism problem in graph terminology.

3 Preliminaries

In this section, we introduce the definition of KGs and the various concepts used in the proposed method.

3.1 Knowledge Graphs

Knowledge graph (KG) is a data structure that represents objects such as people, things, dates, and places as nodes and their relations as labeled directed edges (Kejriwal, Knoblock, and Szekely 2021). Theoretically, a knowledge graph G is defined as

$$G = (V, E, s, t, A, R, \phi, \psi)$$

where V is the set of nodes, E is the set of edges, and the mappings s and $t: E \rightarrow V$ associate each edge $e \in E$ with its source and target nodes, respectively. Additionally, A and R are the sets of node labels and edge labels, and $\phi: V \rightarrow A$ and $\psi: E \rightarrow R$ are mappings that associate nodes and edges with their corresponding labels. In this general definition, the set of nodes V and the set of labels A are distinct. This allows for the assignment of the same label $l \in A$ to different nodes $v_1, v_2 \in V$ ($v_1 \neq v_2$), enabling the system to represent entities with the same name $\phi(v_1) = \phi(v_2)$. In this study, we assume $V = A$ for simplicity, identifying nodes by their label strings such as “Barack Obama” or “Hawaii”. Similarly, we assume $E = R$ and identify each edge $e \in E$ with its label $\psi(e)$.

Focusing on the fact that a knowledge graph G consists of multiple ternary relations (v_1, r, v_2) , where $r \in R$ represents the label of the edge from node v_1 to v_2 , **we represent G as a set of triplets**

$$G = \left\{ (v_1, r, v_2) \mid \begin{array}{l} \exists e \in E \text{ s.t. } v_1 = \phi(s(e)) \\ \wedge r = \psi(e) \\ \wedge v_2 = \phi(t(e)) \end{array} \right\} \\ \subset V \times R \times V.$$

Here, v_1 and $v_2 \in V$ represent the source and target nodes, respectively, and r represents a directed edge $r \in R$.

3.2 Relation Extraction

The task of extracting relations between entities described in a natural language sentence is called **relation extraction (RE)**, and it is an important step to automatically construct KGs (Zhao et al. 2024).

According to Zhao et al. (2024), current RE methods are divided into two categories, namely **pipeline-based approaches** and **joint approaches**. In the setting of pipeline-based approaches, the task of RE is conducted in two stages: (i) named entity recognition (NER) and (ii) relation classification. In the first step, the method finds named entities from the original sentence, for example, if a sentence “Barack Obama, who served as the 44th President of the United States, was born in Hawaii” is given, it extracts entities such as “Barack Obama”, “the 44th President of the United States”, and “Hawaii”. Then, in the second stage, the method classifies relation types between each entity pair found in the NER stage and outputs semantic triplets as the final result. However, such pipeline-based approaches are known to fall short in error propagation, where the error in the first stage affects the performance of the second. In order to overcome this issue, joint RE approaches are proposed to address the task by identifying entities and relation types at the same time.

In this study, we consider two kinds of RE methods that are reported to achieve the best F1 scores in the NYT dataset (Riedel, Yao, and McCallum 2010; Zhao et al. 2024), namely Relation Extraction By End-to-end Language generation (**REBEL**) (Huguet Cabot and Navigli 2021) and Unified Representation and Interaction for Joint Relational Triple Extraction (**UniRel**) (Tang et al. 2022), both of which are categorized into joint RE approaches. Details of the two methods are described in Appendix A. In the following, we denote the relational graph which is created from a natural language sentence x by using a certain RE method as $\mathcal{G}(x)$.

3.3 Entity Linking

Entity Linking (EL) is the task of mapping a textual mention to its corresponding entry in a structured knowledge base (Shen, Wang, and Han 2015). When the target knowledge base is Wikipedia, the task is commonly called Wikification or Disambiguation to Wikipedia (D2W) (Tsai and Roth 2016). Within the field of knowledge graphs, EL is widely used for **entity resolution**, i.e., merging nodes that represent the same real-world entity either within a single KG or across multiple KGs (Sun et al. 2018). Throughout this research, we let $\mathcal{L}$ denote the consecutive operation of applying EL to a knowledge graph G and merging equivalent nodes.

3.4 Semantic Similarity

Semantic textual similarity (STS) is a measure of semantic equivalence between the given two blocks of text (Chandrasekaran and Mago 2021). According to Chandrasekaran and Mago (2021), current STS methods are categorized into (i) knowledge-based methods, (ii) corpus-based methods, (iii) deep neural network-based methods, and (iv) hybrid methods. In this study, we utilize the STS of different node labels as the **similarity of nodes** in KGs. We represent the similarity between the two nodes v_1 and $v_2 \in V$ as $\text{sim}(v_1, v_2) \in \mathbb{R}$.

4 Methodology

In this section, we describe a method for estimating the correct answer in MCQs using an RE method and a KG matching procedure. Our proposing method takes the question sentence $q(x)$ and the set of option words $\mathcal{O}$ as input and estimates the correct option $\hat{o} \in \mathcal{O}$ through the following steps:

1. Create a **propositional graph** $\text{PG}(o_i)$ for each choice o_i to represent the relations extracted from the sentence $q(o_i)$. We also create a **knowledge graph** $\text{KG}(o_i)$ for each choice o_i to represent the background knowledge for solving the question.
2. Estimate the corresponding nodes between $\text{PG}(o_i)$ and $\text{KG}(o_i)$.
3. Verify each triplet in $\text{PG}(o_i)$ against the ones from $\text{KG}(o_i)$ to compute the **truthfulness** of each sentence $q(o_i)$ ($i = 1, 2, 3, 4$).
4. Choose the correct answer based on the truthfulness and the similarity of node labels between $\text{PG}(o_i)$ and $\text{KG}(o_i)$.

Steps 1 to 3 are the procedure for calculating the truthfulness of each proposition $q(o_i)$. Through these steps, fact verification using KG matching is applied to the given $q(o_i)$, and the truthfulness of the proposition being true is computed by predefined rules. The final result is selected in Step 4.

We assume that an RE method $\mathcal{G}$, an entity linking method $\mathcal{L}$, and a semantic similarity measure $\text{sim}(v_1, v_2)$ are predefined. In the following part, this section describes the details of each step.

4.1 Step 1. Creating Propositional Graphs and Knowledge Graphs

Constructing Propositional Graph In the first step, we create a relational graph based on each proposition $q(o_i)$ to represent the relations to be verified. At this time, if o_i is an incorrect option, the relations represented in $q(o_i)$ will also contain factual errors. Therefore, we call the relational graph created in this step, which could contain incorrect relations, a **propositional graph** (PG), and we distinguish it from the **knowledge graph** (KG) that only contains the ground truth triplets. By using an RE method $\mathcal{G}$, we construct such a propositional graph for each choice o_i ($i = 1, 2, 3, 4$) as $\mathcal{G}(q(o_i))$.

In the propositional graph for choice o_i , it is desirable that the structure of graphs other than the option label o_i be isomorphic for other choices. However, when using a machine learning-based RE method, the structure of the graph $\mathcal{G}(q(o_i))$ may differ for each sentence $q(o_i)$ because the method has probabilistic behavior depending on the input. Therefore, to satisfy the aforementioned condition, we create a **propositional graph template** $\text{pg}(\#)$ by (i) replacing the node string o_i in original $\mathcal{G}(q(o_i))$ with a special symbol $\#$ and (ii) joining these $\mathcal{G}(q(\#))$ together:

$$\text{pg}(\#) := \bigcup_{o_i \in \mathcal{O}} \mathcal{G}(q(o_i))\{o_i := \#\},$$

where $\{o_i := \#\}$ means the substitution of $\#$ to o_i . Hereafter, $\text{pg}(o_i)$, which is obtained by substituting o_i for the special symbol $\#$, is used as the propositional graph for each choice, instead of the original $\mathcal{G}(q(o_i))$.

Constructing Knowledge Graph Based on Wikipedia articles, we construct a knowledge graph for verifying the propositional graph. Here, let the set of nodes in $\text{pg}(o_i)$ be V_P . If a human were to verify the proposition $q(o_i)$, they would search for the keywords that are the subject of the sentence on Wikipedia and check if the content of the article matches the claim of the proposition. Our method mimics this process by treating each node label in $\text{pg}(o_i)$ as a topic. Therefore, for each node $v \in V_P$, $\text{label}(v)$ is searched on Wikipedia and the resulting article text $\text{text}(v | o_i)$ is converted into a relational graph.

$$\text{kg}(v | o_i) := \mathcal{G}(\text{text}(v | o_i))$$

$\text{kg}(v | o_i)$ for each node $v \in V_P$ is then joined together to construct the knowledge graph $\text{kg}(o_i)$ representing the

ground truth triplets:

$$\text{kg}(o_i) := \bigcup_{v \in V_P} \text{kg}(v | o_i).$$

Resolving Node Label Ambiguity Because of the notational fluctuations in natural language texts, sometimes multiple nodes in the relational graphs represent the same entity in the real world. In order to resolve such node label ambiguity, we apply the entity linking method for each node in the constructed graphs and replace original labels with entity titles if found. By representing this process with the notation $\mathcal{L}$, the PG and KG for each choice o_i are given as

$$\text{PG}(o_i) = \mathcal{L}(\text{pg}(o_i)), \text{ and}$$

$$\text{KG}(o_i) = \mathcal{L}(\text{kg}(o_i)).$$

4.2 Step 2. Estimating Corresponding Nodes

Let the node sets for $\text{PG}(o_i)$ and $\text{KG}(o_i)$ be V_P and V_K . In the following, we verify the correctness of the proposition $q(o_i)$ by comparing $\text{PG}(o_i)$ and $\text{KG}(o_i)$ created in Section 4.1. The verification is based on the idea that the more edges $(v_1, r, v_2) \in \text{PG}(o_i)$ are included in $\text{KG}(o_i)$, the more likely the original sentence $q(o_i)$ is correct.

However, since these graphs are created by RE, even nodes referring to the same entity may have variations in their labels at this time, depending on the original words in the sentences. Therefore, we need to find the corresponding nodes between the subset of V_P and V_K , namely $V'_P := V_P \setminus V'_K$ and $V'_K := V_K \setminus V_P$, respectively, excluding the nodes with perfectly matched labels. Then, we compute the mapping $\varphi': V'_P \rightarrow V'_K$ that maximizes the sum of sentence similarity between corresponding node labels:

$$\hat{\varphi}' := \arg \max_{\varphi'} \sum_{v \in V'_P} \text{sim}(\text{label}(v), \text{label}(\varphi'(v))).$$

Here, $\varphi': V'_P \rightarrow V'_K$ represents a **bijection** of node correspondences, and $\text{sim}(\cdot)$ represents the sentence similarity between two labels. This problem is a maximum matching problem in a bipartite graph consisting of two node sets V'_P and V'_K . Using the solution $\hat{\varphi}'$, the desired correspondence between the entire node sets $\hat{\varphi}: V_P \rightarrow V_K$ is defined as

$$\hat{\varphi}(v) := \begin{cases} \hat{\varphi}'(v) & \text{if } v \in V'_P, \\ v & \text{if } v \in V_P \cap V_K. \end{cases}$$

4.3 Step 3. Computing Truthfulness

Finally, we count the edges $(v_1, r, v_2) \in \text{PG}(o_i)$ that are also included in $\text{KG}(o_i)$, and define **truthfulness (edge score)** of the original proposition $q(o_i)$ as the ratio of mutually included edges.

We define a projected version of $\text{PG}(o_i)$, namely $\hat{\varphi}(\text{PG}(o_i))$, by replacing the nodes $v_1, v_2 \in V_P$ with the corresponding nodes $\hat{\varphi}(v_1), \hat{\varphi}(v_2) \in V_K$ as

$$\begin{aligned} \hat{\varphi}(\text{PG}(o_i)) &= \{(\hat{\varphi}(v_1), r, \hat{\varphi}(v_2)) \mid (v_1, r, v_2) \in \text{PG}(o_i)\} \\ &\subset V_K \times R \times V_K. \end{aligned}$$

Algorithm 1 Answer selection based on edge and node scores

Input: Lists of edge scores $E = [e_1, \dots, e_4]$ and node scores $N = [n_1, \dots, n_4]$

Output: Index of the best answer

```

1:  $e_{\max} \leftarrow \max(E)$ 
2: if only one element in  $E$  equals  $e_{\max}$  then
3:   return index of  $e_{\max}$  in  $E$ 
4: else
5:    $I \leftarrow \{i \in \{1, 2, 3, 4\} \mid E[i] = e_{\max}\}$ 
6:   Initialize  $N'$  as  $[0, 0, 0, 0]$ 
7:   for all  $i \in I$  do
8:      $N'[i] \leftarrow N[i]$ 
9:   end for
10:   $n_{\max} \leftarrow \max(N')$ 
11:  if only one element of  $N'$  equals  $n_{\max}$  then
12:    return index of  $n_{\max}$  in  $N'$ 
13:  else
14:     $J \leftarrow \{j \in \{1, 2, 3, 4\} \mid N'[j] = n_{\max}\}$ 
15:    Randomly select  $r$  from  $J$ 
16:    return  $r$ 
17:  end if
18: end if

```

Using this notation, the desired truthfulness $\mathcal{T}$ is expressed by the following equation:

$$\mathcal{T}(o_i) := \frac{|\hat{\varphi}(\text{PG}(o_i)) \cap \text{KG}(o_i)|}{|\text{PG}(o_i)|} \in [0, 1]$$

Here, $|G|$ represents the number of semantic triplets in a relational graph G .

With this $\mathcal{T}$, we consider that the propositional graph $\text{PG}(o_i)$ is proved by the knowledge graph $\text{KG}(o_i)$ with the confidence of $\mathcal{T}(o_i)$ and denote it as

$$\text{KG}(o_i) \vdash_{\mathcal{T}(o_i)} \text{PG}(o_i).$$

In addition, we regard the original proposition $q(o_i)$ is modeled by the ground truth knowledge C_i retrieved from Wikipedia with the confidence of $\mathcal{T}(o_i)$, and denote it as

$$C_i \models_{\mathcal{T}(o_i)} q(o_i).$$

4.4 Step 4. Estimating Correct Option

By applying the above three steps to $q(o_i)$ ($i = 1, 2, 3, 4$), we estimate the correct option $\hat{o}$ as the one with the maximum truthfulness $\mathcal{T}(o_i)$:

$$\hat{o} := \arg \max_{o_i \in \mathcal{O}} \mathcal{T}(o_i).$$

If there are multiple o_i that give the maximum value for $\mathcal{T}(o_i)$, we also take the average label similarity of the corresponding nodes $\mathcal{N}(o_i)$ into account in addition to $\mathcal{T}(o_i)$.

$$\mathcal{N}(o_i) := \frac{1}{|V_P|} \sum_{v \in V_P} \text{sim}(\text{label}(v), \text{label}(\hat{\varphi}(v)))$$

In the following part, we call this $\mathcal{N}(o_i)$ the **node score**. If $\hat{o}$ cannot be narrowed down to one, the method selects randomly from the choices with the highest node score. The overall algorithm for selecting the best answer is summarized in Algorithm 1.

Category Name	Questions
Art & Music	20
General Knowledge	20
Geography	20
History	20
Literature & Language	20
Mathematics	20
Philosophy & Logic	20
Pop Culture	20
Science	20
Technology & Computing	20
Total	200

Table 1: 10 question categories of KR-200m and KR-200s dataset.

5 Experiments

In this section, we conduct answering experiments on original MCQ datasets to demonstrate the effectiveness and the traceability of our method. We also compare the performances of various RE methods over different lengths of question sentences to investigate the method’s capability on each question category. In addition, we conduct an ablation study to show the importance of entity linking in the graph creation process.

5.1 Experimental Setup

MCQ Datasets For the answering experiment, we created two variants of the MCQ dataset in the “fill-in-the-blank” format, namely KR-200m and KR-200s, using GPT-4o (OpenAI 2024), both of which consisted of 200 questions over 10 categories shown in Table 1. The difference between KR-200m and KR-200s is the length of the question sentence $q(x)$. KR-200m has 20.1 words in $q(x)$ on average, including blank position x counted as one word, while KR-200s has only 7.5 words per sentence.

Relation Extraction Methods For the used RE methods, we consider REBEL (Huguet Cabot and Navigli 2021), mREBEL₄₀₀, mREBEL₃₂ (Huguet Cabot et al. 2023), and UniRel (Tang et al. 2022) trained on the NYT dataset (Riedel, Yao, and McCallum 2010). The details of each RE method are described in Appendix A.

Entity Linking Method For the simplicity of the implementation, we construct an entity linking (EL) method using the title search of the Wikipedia API. In this method, we search for the original node labels in Wikipedia and link them to the article’s title, which the API returns as the most relevant. If no page is found, we do not assign any page title to the node, leaving the original node label.

Semantic Similarity Measure For the semantic similarity measure, we use sentence embedding model all-MiniLM-

	EL Setting	KR-200m	KR-200s
REBEL	w/ EL	53.5	48.0
	-	42.2	42.0
mREBEL ₄₀₀	w/ EL	52.2	49.7
	-	43.3	41.6
mREBEL ₃₂	w/ EL	49.0	43.6
	-	43.3	37.7
UniRel	w/ EL	28.6	26.9
	-	29.0	27.3

Table 2: Overall results of answering experiments by different RE methods, with and without entity linking (EL) methods applied in graph creation. The best results across different datasets and settings are highlighted in **bold**. Accuracy (%) is used as the evaluation metric.

L6-v2¹, which converts a sentence s into a 384-dimensional embedding vector, and define the semantic textual similarity $\text{sim}(v_1, v_2)$ between sentences v_1 and v_2 as the cosine similarity of their embedding vectors. all-MiniLM-L6-v2 is a model trained on a dataset of over one billion sentences, based on the lightweight language model MiniLM (Wang et al. 2020).

Implementation Details For the implementation of REBEL and mREBEL series, we set the max length of generated sequences to 256, the length penalty to 0, and the number of beams and the number of return sequences to 3 in the settings of the beam search algorithm to realize the diverse representation of the output graph.

All Wikipedia pages are fetched on July 9th, 2025, via MediaWiki API endpoint². For creating KGs, we used the summary part of Wikipedia articles, which appears before any sections on the page.

We conducted the experiments using an NVIDIA RTX 5000 Ada GPU with 32 GB of RAM. The source code for the experiments and the MCQ datasets are available at our GitHub repository³.

5.2 Results

Overall Accuracy The overall results of answering experiments across different RE methods and EL settings are shown in Table 2. According to the table, REBEL and its variant mREBEL₄₀₀ achieve the highest accuracy on the two datasets, offering 53.5% and 49.7% accuracy on KR-200m and KR-200s, respectively. Following the two methods, mREBEL₃₂ performs at slightly lower accuracy, while UniRel merely answers with as much accuracy as a random chance. These results show that each model’s number of relation types has an essential effect on accuracy. However, despite the vast difference between the number of relation types that REBEL and mREBEL₄₀₀ can generate (220 and

¹<https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2>

²https://www.mediawiki.org/wiki/API:Main_page

³<https://github.com/N-Shimoda/KG-MCQA>

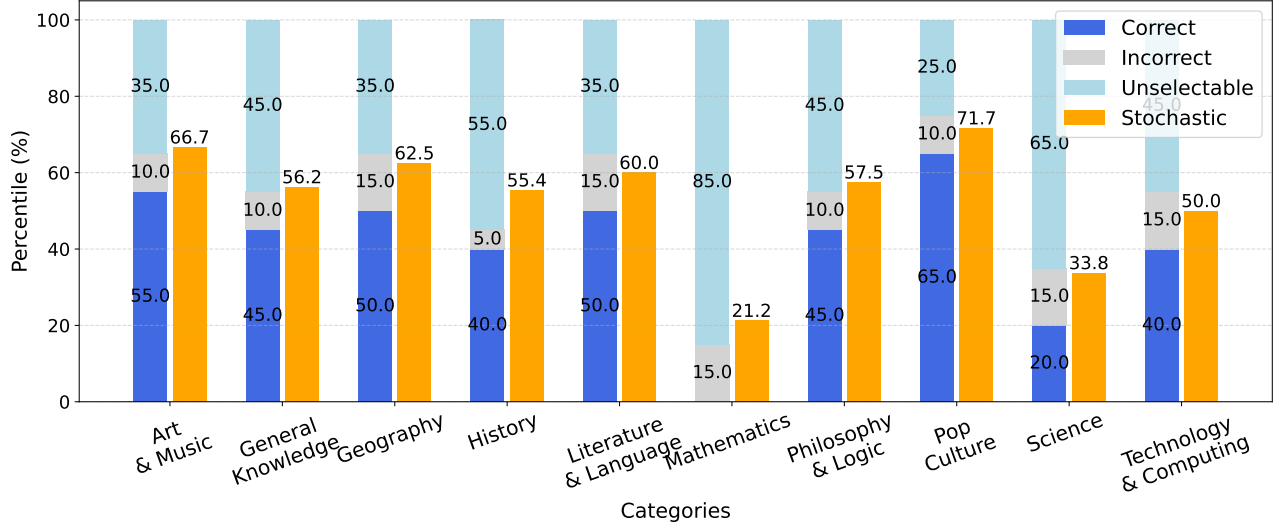


Figure 1: Accuracy per category for REBEL with entity linking on KR-200m. The columns with *Correct*, *Incorrect*, and *Unselectable* show the results without random choice, determined solely by the edge and node scores. *Stochastic* shows the accuracy with random selection for the unselectable questions (Section 4.4).

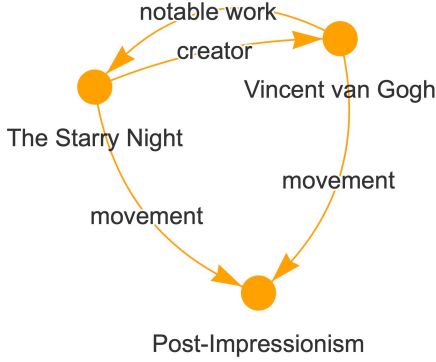


Figure 2: PG for the **correct** option o_1 : “Starry Night”.

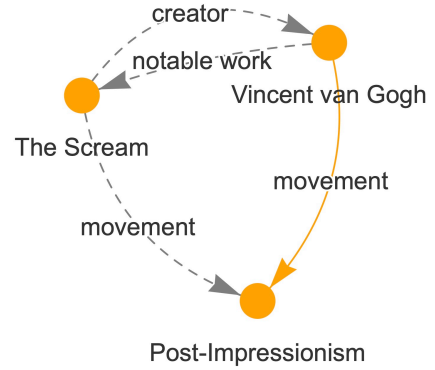


Figure 3: PG for the **incorrect** option o_1 : “The Scream”.

400, respectively), both models perform almost equally on the two datasets, indicating that too large a number of relation types does not improve the result anymore. In addition, entity linking enhances the accuracy of the method for REBEL and mREBEL series, exhibiting the importance of node label disambiguation in the graph creation process.

In addition, we show the performance comparison of our method and the baseline open-source large language model in Appendix B.

Results by Category Figure 1 shows the accuracy per category for REBEL with EL enabled, which performed the best score on the KR-200m dataset. Among the 10 categories, our method achieved the highest accuracy of 71.7% on “Pop Culture”, followed by “Art & Music”, “General Knowledge”, “Geography”, “History”, “Literature & Language”, “Philosophy & Logic”, and “Technology & Computing” with 50.0% to 66.7% accuracy. In contrast, the

method did not perform well in two categories: “Mathematics” and “Science”. These results indicate that our method achieves higher accuracy in categories where factual knowledge, represented as semantic triplets, is required to select the best option. On the other hand, it does not perform well on questions that require abstract knowledge or logical inferences, such as mathematical derivations.

5.3 Case Study of Traceability

For a more intuitive understanding of the results, we illustrate PGs and KGs for an MCQ. The question is from KR-200m, “Art & Music” category, and the question sentence $q(x)$ and the set of option words $\mathcal{O}$ are shown in Example 2.

Example 2 (Starry Night). Let $q(x)$ be “Vincent van Gogh, a Dutch post-impressionist painter, created several master-

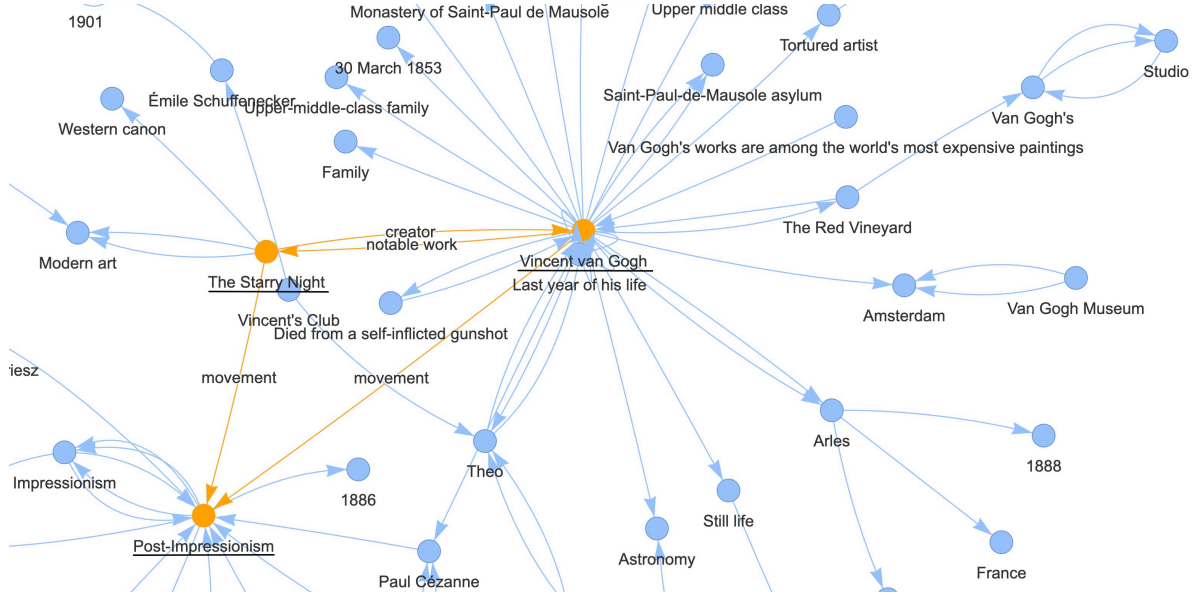


Figure 4: Excerpt of the KG used for verifying PG of the **correct** option o_1 : “Starry Night”. Nodes with underlined labels correspond to the PG nodes.

pieces, including the famous painting called $\{x\}$,” where

$$\mathcal{O} = \left\{ \begin{array}{l} o_1: \text{“Starry Night”}, \\ o_2: \text{“The Persistence of Memory”}, \\ o_3: \text{“Guernica”}, \\ o_4: \text{“The Scream”} \end{array} \right\}.$$

The correct answer is o_1 : “Starry Night”.

Figure 2 and 3 show the PGs for the correct option o_1 : “Starry Night” and the incorrect option o_4 : “The Scream”, respectively. Edges shown as solid lines are the ones verified from the KGs. Comparing these two graphs, we can understand that the correct option o_1 : “Starry Night” was selected because all relations in $\text{PG}(o_1)$ are verified, whereas only one edge from “Vincent van Gogh” to “Post-Impressionism” is verified for $\text{PG}(o_4)$. Through such a comparison between the PGs for the chosen option and the one not selected, our method offers traceability on why a certain choice was selected or **not** selected.

In addition to the interpretability of the difference between two options, our method offers explainability for the rationale behind the calculated truthfulness. Figure 4 visualizes an excerpt of $\text{KG}(o_1)$, which was used for verifying $\text{PG}(o_1)$ for the **correct** option o_1 . By visualizing such KGs, we can understand what kind of relations are extracted from given Wikipedia articles and which semantic triplets are used for the verification.

6 Related Works

In the proposed method, we handle MCQs in the cloze test format (i.e., “fill-in-the-blank” format) and select answers by comparing the **truthfulness** of sentences with the options inserted into the question sentences (Section 4). In this section, we explain previous research on **multiple choice**

question answering (MCQA) and **fact verification**, which predicts if a given sentence is factually correct or not.

6.1 MCQ with DNNs

MCQA is a task where a model is applied to select the correct option given a question sentence q , a set of options $\mathcal{O} = \{o_1, \dots, o_k\}$, and optional background knowledge $\mathcal{C} = \{C_1, \dots, C_k\}$ related to each choice o_i ($i = 1, \dots, k$) (Shah, Gupta, and Roth 2020). Existing datasets include MMLU (Hendrycks et al. 2021) and BIG-bench (BIG-bench authors 2023), intended for evaluating the performance of LLMs. Therefore, most methods tackling MCQA naturally involve LLMs. On the other hand, a combination of KGs and language models has been proposed to enhance reasoning capabilities. Zhang et al. (2023) proposed a method that retrieves subgraphs from a knowledge graph database tailored for the question sentence q and the set of options $\mathcal{O}$ and inputs them into a Graph Neural Network (GNN) to obtain answers. This method is similar to our proposed method in that it creates KGs tailored to the problem setting. However, it differs from our method because it targets question formats other than cloze tests and utilizes language models and reasoning mechanisms. Additionally, since it uses GNNs, there is a problem of low explainability regarding the process until the output results are obtained.

6.2 Fact Verification

Fact verification is a task that determines whether a natural language sentence is factually correct, also known as fact-checking (Guo, Schlichtkrull, and Vlachos 2022). Although fact verification is a natural language task, language models lack clarity in the process leading to the answer, resulting in low explainability. Therefore, methods using KGs with explicit link structures have been proposed. Yuan

and Vlachos (2024) proposed a method that extracts semantic triplets (subject, relation, object) from sentences of unknown veracity and verifies them against a knowledge graph database. This study aligns with our method in converting sentences to be verified into a list of triplets. On the other hand, the method for preparing the ground truth knowledge graph differs from ours, and they utilized additional reasoning modules.

Tackling MCQs on the cloze test format is positioned as a special case of MCQA. However, in MCQA, many previous studies use LLMs without the interest of interpretability, and methods pursuing traceability are in the minority. Therefore, in this study, we propose an explainable MCQA method using KG-based fact verification with considering the case of the cloze test format.

7 Conclusion

This paper presents a combination of the RE method and KG matching for answering MCQs in the “fill-in-the-blank” format. In our framework, the sentence, which is formed by substituting a choice word into the blank position of the question, is regarded as a proposition. We proposed a fact verification method by converting the sentence into a relational graph and verifying it against a factually correct KG. We conducted the answering experiments using various RE methods in two original MCQ datasets with different question lengths. Our method was able to answer up to 70% of the questions, depending on the category, while keeping the traceability of the chosen answer by visualizing verified edges in propositional graphs and the knowledge graph triplets used for verification. The result comparison over different RE methods reveals the tendency for a greater number of relation types in the RE method to cause higher accuracy. In addition, the ablation study demonstrates the importance of the entity linking method in the graph creation process. Future research could improve the accuracy of article search in the KG creation process in order to prevent the mislinking of Wikipedia pages and introduce a more flexible verification mechanism to implement reasoning capability in the method.

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Code and Data Availability

The source code and dataset used in this study are publicly available at: <https://github.com/N-Shimoda/KG-MCQA>.

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A Details of Relation Extraction Methods

In this section, we describe the details of the relation extraction (RE) methods used in this study. The training dataset and the number of relation types for each RE method are summarized in Table 3.

REBEL Series In the setting of REBEL, relation extraction (RE) is conducted as a translation task where the input is a natural language sentence x and the output is a sequence y that represents a list of semantic triplets extracted from x (Huguet Cabot and Navigli 2021). It utilizes three types of special tokens, namely `<triplet>`, `<subj>`, and `<obj>`, to linearize semantic triplets into a sequence y . For example, a triplet (“Barack Obama”, “born_in”, “Hawaii”) is linealized as follows.

```
<triplet> Barack Obama <subj> Hawaii
<obj> born_in
```

The REBEL model has an architecture based on BART-large (Lewis et al. 2020), an encoder-decoder Transformer (Vaswani et al. 2017), and is designed to generate 220 relation types.

In addition to the original model, we utilize two multilingual variants of REBEL (mREBEL₄₀₀ and mREBEL₃₂), which are based on the mBART-50 architecture (Tang et al. 2021). mREBEL₄₀₀ is trained on automatically annotated dataset named RED^{FM} which includes 400 relation types, whereas mREBEL₃₂ is trained on SRED^{FM} dataset which is created from RED^{FM} via human supervision, including 32 relation types (Huguet Cabot et al. 2023).

Model	Training Dataset	Relation Types
REBEL	REBEL dataset	220
mREBEL ₄₀₀	RED ^{FM}	400
mREBEL ₃₂	SRED ^{FM}	32
UniRel	NYT	24

Table 3: Training dataset and number of relation types for each RE method used in this study.

Method	KR-200m	KR-200s	P-100
ours (REBEL)	53.5	48.0	62.3
ours (mREBEL ₄₀₀)	52.2	49.7	67.3
FLAN-T5-xlarge	68.0	61.0	55.0

Table 4: Results of answering experiments by our method and the open-source LLM. The best results across different datasets are highlighted in **bold**. Accuracy (%) is used as the evaluation metric.

UniRel UniRel (Tang et al. 2022) reformulated RE as a joint tagging problem in which entities and relations are encoded in a single natural-language sequence, allowing the model to reason over their mutual dependencies at once. Concretely, a sentence x is concatenated with every relation label expressed as an English phrase (e.g., “born in”, “capital of”), and this long sequence is fed into a BERT-base encoder (Devlin et al. 2019). From the self-attention weights in the last layer, UniRel builds an **Interaction Map**, a sigmoid-scaled matrix whose cells represent two kinds of links: (i) E-R interactions that represent entity-relation links, and (ii) E-E interactions that indicate subject-object links. Since both interaction types share the same map, the model learns unified patterns instead of two heterogeneous classifiers, which both improve F1 and cut inference time to more than 2.6 times faster than previous joint methods.

B Comparison with LLM Baseline

In this section, we compare the performance of our method with an open-source large language model (LLM) on crafted MCQ datasets. For the LLM, we consider **FLAN-T5-xlarge** (Chung et al. 2024) with 3B parameters. For the MCQs, in addition to the two datasets introduced in Section 5, we also use **P-100 dataset** consisting of 100 questions about people generated by GPT-4o (OpenAI 2024). We created the P-100 dataset in order to validate the empirical assumption from Figure 1 that our method could have a tendency to perform well on questions about people, such as the ones in “Pop Culture”, where the relations between the person and the related concepts are the main focus of the question.

The results of answering experiments by our method and FLAN-T5-xlarge are shown in Table 4. For KR-200m and KR-200s, FLAN-T5-xlarge performed better than our method, whereas ours using mREBEL₄₀₀ performed the best on the P-100 dataset with exceeding FLAN-T5-xlarge by 12.3 percentage points. These results highlight the superior accuracy of LLMs in the MCQ datasets consisting of various categories. However, it is also noticeable that our method could perform better on certain types of questions than zero-shot LLM while keeping the interpretability of the answering process.